

USBasp wiring and firmware upload

Six wires. Every pin you need is on the board's top row. Nothing else connected: no FTDI, no battery, no string.

1 The USBasp 10-pin header, looking at the connector

odd row: 1 3 5 7 9

1 MOSI	3 NC	5 RST	7 SCK	9 MISO
2 VCC	4 GND	6 GND	8 GND	10 GND

Pin 1 is marked with a dot, a triangle or a square pad. On a ribbon cable it is the red stripe. Only 1, 2, 5, 7, 9 and one GND are used.

even row: 2 4 6 8 10

2 The Pro Mini top row, board the right way up, FTDI header on the left

RAW	GND	RST	VCC	A3	A2	A1	A0	13	12	11	10
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Coloured pins are the six you use. A4 to A7 are the unpopulated holes at the far end; ignore them.

3 The six connections

USBasp	Signal	Pro Mini pin	Suggested wire
Pin 1	MOSI	11	green
Pin 9	MISO	12	blue
Pin 7	SCK	13	yellow
Pin 5	RESET	RST	orange
Pin 2	VCC	VCC	red
Pin 4, 6, 8 or 10	GND	GND	black

RESET goes to RST, not to a numbered pin. MOSI and MISO are the pair people swap: 11 out, 12 back.

4 USBasp setup, then the commands

Set the fuses once on a new board (docs/08-firmware.md). Routine uploads never rewrite them.

Use a verified 3.3 V target power and logic setting for the isolated controller.

Some clone voltage jumpers change target power but not signal level: verify the actual unit with a meter.

JP2 is only for updating the USBasp's own firmware. Leave it open.

JP3, "slow SCK": fit it if the chip will not answer. Old clone firmware ignores avrdude's -B option.

Plug in the USBasp (Windows needs a libusb driver; macOS and Linux do not). Test first, this writes nothing:

avrdude -c usbasp -p m328p -v

Expect: Device signature = 1E 95 0F (ATmega328P). Stop on all 00 / all ff or anything else, except:

1E 95 16 (ATmega328PB): same firmware and fuses; use -p m328pb in every command. Never use -F.

From firmware/controller, compile the advanced build with:

pio run -e advanced

Then flash with the separately installed current avrdude, not pio -t upload:

avrdude -c usbasp -p m328p -v -U flash:w:.pio/build/advanced/firmware.hex:i

For deployment use -e production and .pio/build/production/firmware.hex. Check the verify step passes.

5 If it will not talk

Message	Usual cause
could not find USB device 0x16c0/0x5dc	USBasp not enumerated: cable, hub, or a dead clone
device signature = 0x000000	No power at the target, or a loose wire. Check VCC on the board
signature reads as 0xfffff	MISO not connected, or MOSI and MISO swapped
signature 1E 95 16 (ATmega328PB)	Not a fault: use -p m328pb instead of -p m328p
initialization failed, rc=-1	SCK too fast: fit JP3, or try -B 8
warning: cannot set sck period	Harmless on old clone firmware. Use JP3 instead

Never add fuse-write options to a routine application upload. Disconnect ISP before restoring battery power.